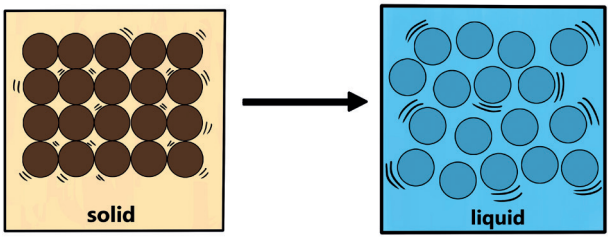
The matter begins to expand when heated and the volume increases due to the increase in the distance between the particles. But, the size of the particles remains same.

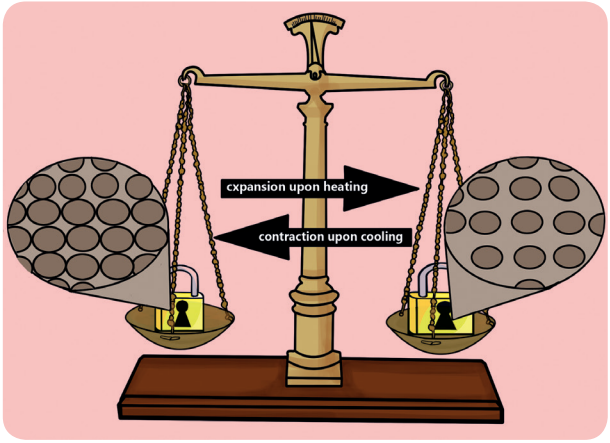




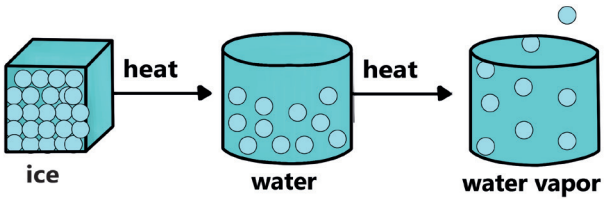
How do hot-air balloons float? When air inside the hot air balloon is heated with a burner, it expands. The expansion causes the density of the air inside the balloon to decrease. Hence, the air inside the balloon has a lower density than the air outside the balloon. This difference in density allows the hot-air balloon to float.



During heating or expansion, the mass of matter does not change. Although the volume of the matter changes, the size and number of the particles of matter do not change. Hence, during heating, the mass of matter is conserved. For example, in an iron lock the distance between the iron particles increases when they gain enough heat. However, the number of iron particles does not change. Hence, the mass of the iron lock is conserved.



The melting of ice is an example for change of states of matter. The change in the states of matter occurs during melting, boiling and freezing and condensation. When the particles possess enough energy, they overcome the strong forces of attraction between one another. They break free from one another and move randomly. For example, when solid ice is heated to 0^0C, it melts to become liquid water. In the same way, when liquid water is heated to 100^0C, it boils to become steam.



1. Solid
When solid is heated, the particles gain energy and vibrate more vigorously
2. Liquid
When the melting point is reached melting occurs. The solid changes to its liquid state.
When a liquid is heated the particles gain energy and vibrate more vigorously.
3. Gas
Boiling occurs when the boiling point is reached. The liquid changes to its gaseous state.